

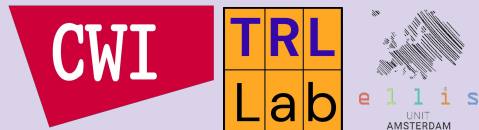
On Different Flavors hence Needs for *Tabular Reasoning*

Madelon Hulsebos

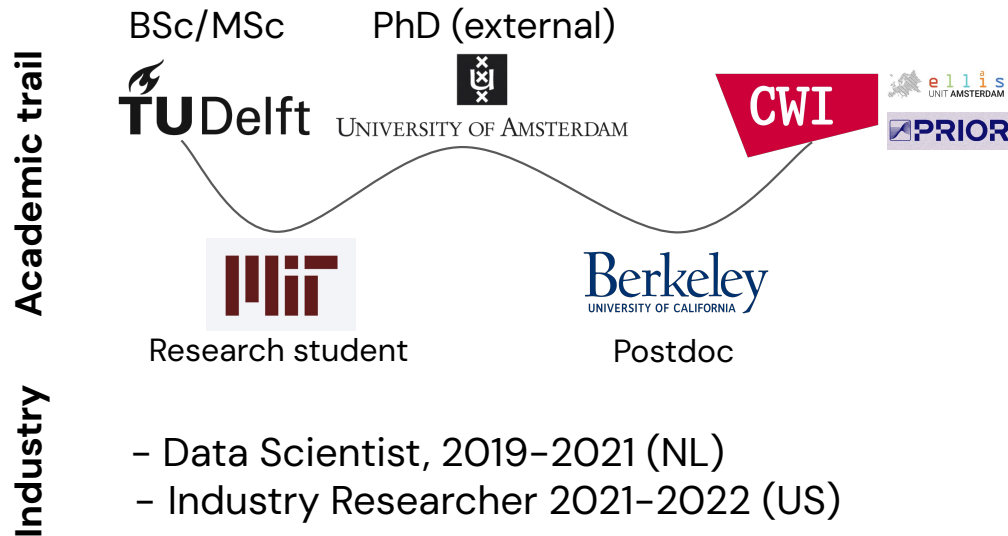
25 September 2026

BIFOLD Colloquium

TU Berlin



Madelon Hulsebos



madelonhulsebos.com
trl-lab.github.io

Current:

- **Tenured Researcher** at CWI, leading the Tabular Reasoning & Learning (TRL) Lab*
- **Faculty & MT Member** ELLIS unit Amsterdam
- **Scientific Advisor** Prior Labs (🇩🇪 Tabular AI startup) *founded TRL workshop series @neurips/acl in '22

TRL Lab @ CWI

Research vision:

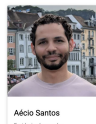
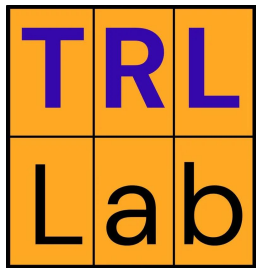
prevailing

actionable

critical

Democratizing insights from tabular data

- 1) **DS/DA from 1 to 10**; so data scientists can focus on things that matter
- 2) **everyone else from 0 to 1**; democratize insights from data



Adão Santos

Agentic systems
for relationship
discovery



Xue (Efty) Li

Knowledge-
grounded agentic
data science



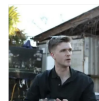
Cornelius Wolff

Tabular
insight
generation



Daniel Gamm

End-to-end
tabular QA



Dennis Loeble

Multi-modal
tabular learning
(predictive)

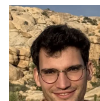


Rohith Prabhakaran

Scaling tabular
foundation
models



Daniel De Dios Alague



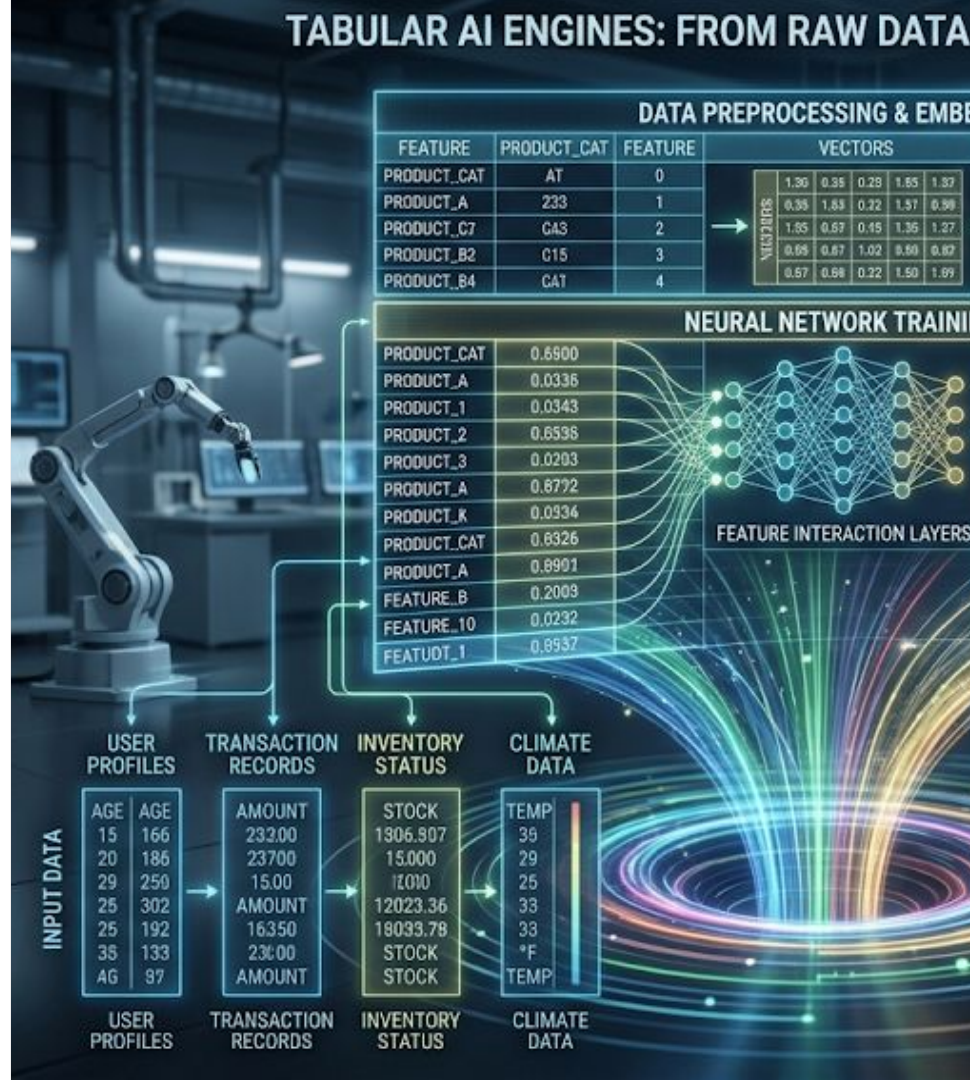
Félix Lefebvre

Context-rich
(pred) tabular
learning

incoming

<https://trl-lab.github.io/>

The case for Tabular AI



AI has made big leaps in...

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas porttitor con Fusce posuere, magna sed pulvinar u lectus malesuada libero, sit amet com eros quis urna. Nunc viverra imperdi est. Vivamus a tellus. Pellentesque ha tristique senectus et netus et malesua turpis egestas. Proin pharetra nonum Mauris et orci. Aenean nec lorem. In p Donec laoreet nonummy augue. Susp

Text



Images

```
sentiments  write_sql.go  parse_expenses.py  addresses.rb

1  #!/usr/bin/env ts-node
2
3  import { fetch } from "fetch-h2";
4
5  // Determine whether the sentiment of text is positive
6  // Use a web service
7  async function isPositive(text: string): Promise<boolean> {
8    const response = await fetch('http://text-processing.com/api/sentiment/', {
9      method: "POST",
10     body: `text=${text}`,
11     headers: {
12       "Content-Type": "application/x-www-form-urlencoded",
13     },
14   });
15   const json = await response.json();
16   return json.label === "pos";
17 }
```

Code...

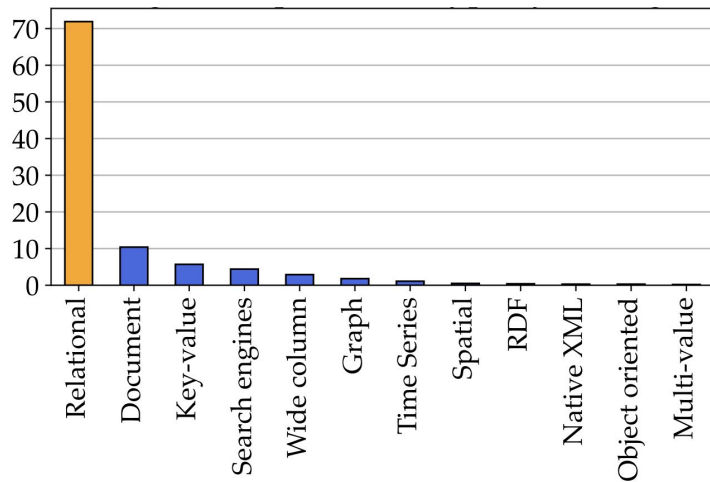
crop rotation : Tabelle													
	Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
	1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
	2	68		100	winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
	3	68		97	winter wheat		sugar beets	fallow land (5,5y)		sandy loam, loe	636	9,6	1993-1996
	4	136		95	winter wheat		oats	sugar beets		sandy loam, loe	636	9,6	
	5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
	6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
	7	136		107	winter wheat		sugar beetsn	summer wheat	maize	sandy loam, loe	636	9,5	
	8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
	9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
	10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
	11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
	12	57 371		98	winter wheat	Sperber	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
	13	57 365		98	winter wheat	Sperber	potatos	sugar beets	summer barle	sandy loam, loe	635		cultivation, weed
	14	57 365		105	winter wheat	Sperber	sugar beets	maize	maize	sandy loam, loe	635		1987-1992
	15	57 365		97	winter wheat	Sperber	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
	16	39 433		90	winter wheat	Okapi	summer barley			sandy loam, loe	690	8,5	oats, cultivation, weec
	17	39 433		100	winter wheat	Okapi	oats			clay, silt	690	8,5	1982-1986
	18	39 433		97	winter wheat	Okapi	winter wheat			clay, silt	690	8,5	

Tables?

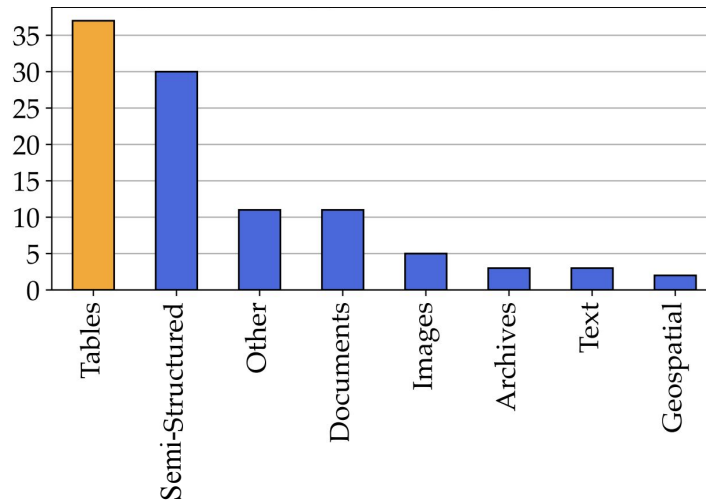
Why?

1 tables **prevail** in organizations

Popularity of database systems



Distribution Google Dataset Search



2 tables serve high value use-cases

Patient ID	Age range	MS age onset	OB	Pregnancy	Lesional load (YYYY/MM-number)	Spinal lesions (YYYY/MM-number)	Treatment
1070	20-29	20	yes	no	2007/03-32007/06-5	2007/09: IFN-β1a	
					2010/10-2010/12-2013/05-2015/06-		
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
3736	30-39	30	yes	yes, from 2015/11 to 2016/08	2014/09-2015/01-		
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
2789	30-39	33	yes	yes, from 2012/10 to 2013/07 and from 2015/11 to 2016/08	2013/11-2013/12-2014/09-2017/01-		
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR
					Displacement due to Conflict in Ukraine (Registration)	Ukraine	UKR

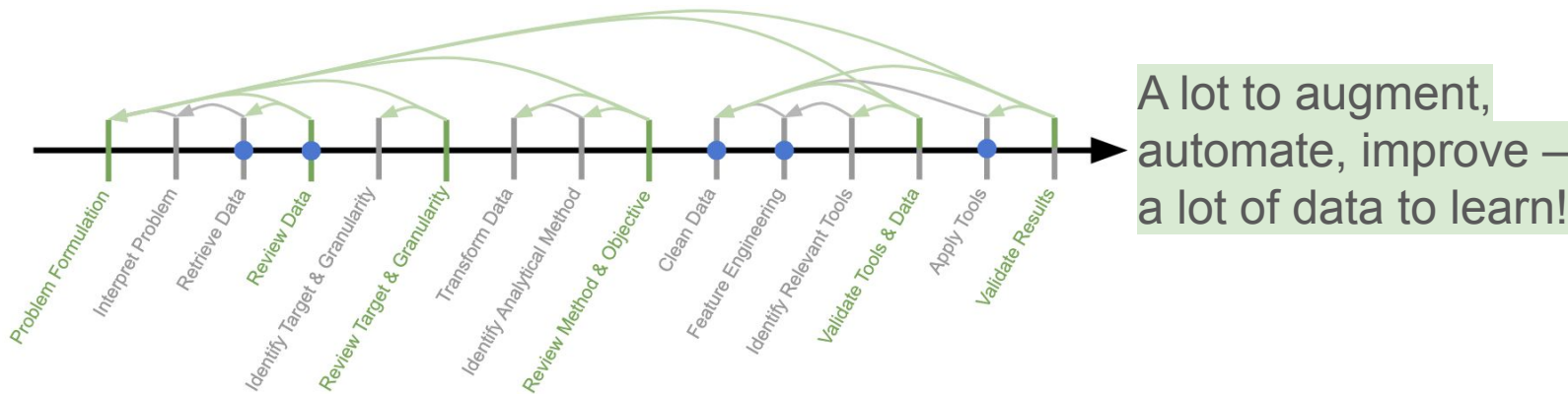
HEALTH

PEACE

	displacementReason	numberMales	numberFemales	idpOriginAdmin1Name								
10	Conflict		67	126	Zaporizka							
20	Conflict		99	156	Odeska							
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2020												
2030												
204												

ECONOMY

3 repetitive, multifaceted, multi-step pipelines



“Position: Unlocking the Full Potential of Data Science Requires Tabular Foundation Models, Agents, and Humans” by Cong, et al., 2025

The Challenges of Tabular AI ❤️



Challenge 2: Data Heterogeneity

Tables come w/ mixed **data types**, **hierarchies**, **semantics** and **sizes**...

Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68	91		winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
2	68	100		winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
3	68	97		winter wheat		sugar beets	fallow land (5.5y)		sandy loam, loe	636	9,6	1993-1996
4	136	95		winter wheat		potatos	sugar beets		sandy loam, loe	636	9,6	
5	136	96		winter wheat		sugar beets	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136	107		winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
7	136	107		winter wheat		sugar beets	summer wheat	maize	sandy loam, loe	636	9,5	
8	136	82		winter wheat		potatos	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136	77		winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136	85		winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136	84		winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
12	57_371	98		winter wheat	Sperber	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
13	57_365	98		winter wheat	Sperber	potatos	sugar beets	summer bark	sandy loam, loe	635		cultivation, weed
14	57_365	105		winter wheat	Sperber	sugar beets	maize	maize	sandy loam, loe	635		1987-1992
15	57_365	97		winter wheat	Sperber	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
16	39_433	90		winter wheat	Okapi	summer barley			sandy loam, loe	690	8,5	oats, cultivation, weec
17	39_433	100		winter wheat	Okapi	oats			clay, silt	690	8,5	1982-1996
18	39_433	97		winter wheat	Okapi	winter wheat			clay, silt	690	8,5	

	2019 Profit	Quantity	Sales	delta Quantity	Sales	Profit
Overall	128.9k	13.3k	1.0m	16.5%	16.2%	36.9%
France	35.1k	3.9k	308.4k	33.8%	33.2%	8.7%
Austria	7.5k	299.0	24.6k	13.3%	11.9%	40.8%
Belgium	4.2k	202.0	17.3k	33.2%	32.7%	36.5%
Denmark	1.3k	96.0	2.8k	1.9%	21.1%	22.9%
Zealand	245.0	15.0	242.7			
South Denmark	362.9	40.0	1.3k	33.3%	7.7%	41.1%
Sonderborg	45.4	6.0	87.5			
Odense	280.3	30.0	1.1k	0.0%	20.1%	54.1%
Esbjerg	37.3	4.0	88.7			
Hovedstaden	0.7k	31.0	1.3k	40.4%	40.9%	35.4%
Copenhagen	0.7k	31.0	1.3k	0.0%	29.5%	23.7%
Frederiksberg	232.1	11.0	1.1k			
Finland	36.0k	2.4k	216.5k	79.8%	61.4%	88.9%
Germany	3.9k	152.0	7.2k	15.0%	46.3%	33.9%
Ireland	10.6k	1.4k	109.7k	17.8%	101.6%	37.8%
Italy	11.6k	0.6k	23.7k	37.5%	37.2%	2.8%
Netherlands	2.9k	135.0	12.9k	46.3%	220.8%	140.8%
Norway	1.0k	74.0	1.8k	29.5%	79.2%	32.4%
Portugal	20.6k	1.2k	99.1k	46.7%	12.9%	27.1%
Spain	9.4k	360.0	15.6k	143.2%	167.7%	128.7%
Sweden	2.2k	84.0	7.3k	8.0%	25.6%	26.1%
Switzerland	36.8k	2.4k	194.0k	25.4%	26.7%	32.7%
United Kingdom	1.3k	48.0	4.0k	30.2%	30.7%	26.7%
Wales	1.9k	118.0	6.6k	5.9%	54.8%	55.5%
Scotland	33.6k	2.2k	183.4k	0.9%		

Cong, Hulsebos, et al., *Observatory: Characterizing Relational Table Embeddings*. VLDB, 2023.

Van Breugel & Van der Schaar, *Position: Why Tabular Foundation Models Should Be a Research Priority*. ICML, 2024.

The task



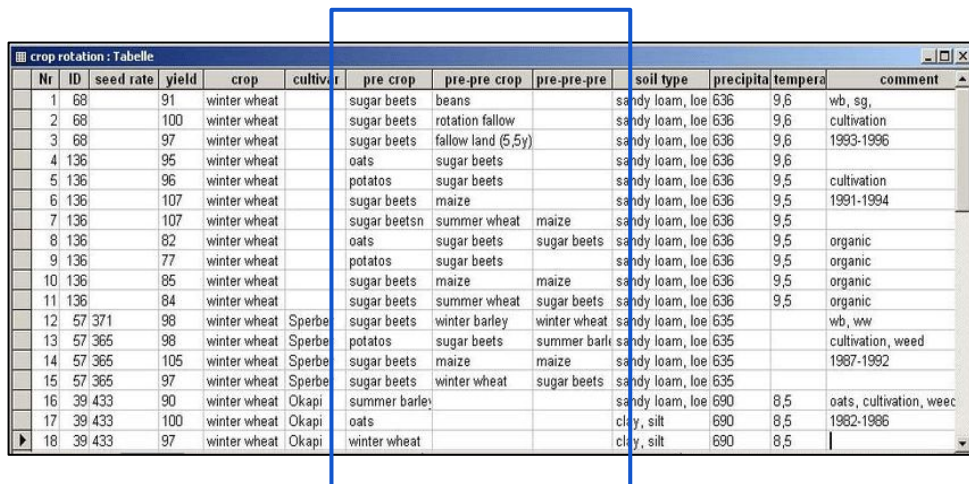
Tabular Reasoning?

Reasoning is the making or granting of assumptions called premises (**starting points**) and the process of moving toward conclusions (end points) from these assumptions by means of warrants – Walton's first pass in "What is Reasoning? What is an Argument?"

Today different "flavors" of reasoning: semantic, analytical, and inferential

Back to the Basics: Semantic Reasoning

What is this table about?



Nr	ID	seed rate	yield	crop	cultivar	pre crop	pre-pre crop	pre-pre-pre	soil type	precipita	tempera	comment
1	68		91	winter wheat		sugar beets	beans		sandy loam, loe	636	9,6	wb, sg,
2	68		100	winter wheat		sugar beets	rotation fallow		sandy loam, loe	636	9,6	cultivation
3	68		97	winter wheat		sugar beets	fallow land (5.5y)		sandy loam, loe	636	9,6	1993-1996
4	136		95	winter wheat		oats	sugar beets		sandy loam, loe	636	9,6	
5	136		96	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	cultivation
6	136		107	winter wheat		sugar beets	maize		sandy loam, loe	636	9,5	1991-1994
7	136		107	winter wheat		sugar beetsn	summer wheat	maize	sandy loam, loe	636	9,5	
8	136		82	winter wheat		oats	sugar beets	sugar beets	sandy loam, loe	636	9,5	organic
9	136		77	winter wheat		potatos	sugar beets		sandy loam, loe	636	9,5	organic
10	136		85	winter wheat		sugar beets	maize	maize	sandy loam, loe	636	9,5	organic
11	136		84	winter wheat		sugar beets	summer wheat	sugar beets	sandy loam, loe	636	9,5	organic
12	57 371		98	winter wheat	Sperbe	sugar beets	winter barley	winter wheat	sandy loam, loe	635		wb, ww
13	57 365		98	winter wheat	Sperbe	potatos	sugar beets	summer barl	sandy loam, loe	635		cultivation, weed
14	57 365		105	winter wheat	Sperbe	sugar beets	maize		sandy loam, loe	635		1987-1992
15	57 365		97	winter wheat	Sperbe	sugar beets	winter wheat	sugar beets	sandy loam, loe	635		
16	39 433		90	winter wheat	Okapi	summer barl			sandy loam, loe	690	8,5	oats, cultivation, weec
17	39 433		100	winter wheat	Okapi	oats			clay, silt	690	8,5	1982-1986
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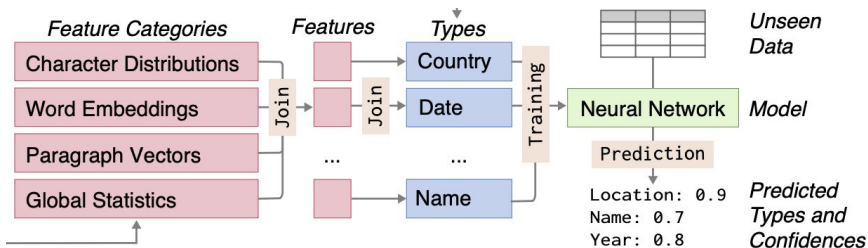
Pre crop?
Pre-pre crop?
Pre-pre-pre?

Semantics key in downstream tasks (data cleaning, integration, predictive ML)

Can we infer *column semantics* w/ embeddings?

Sherlock: neural model w/ thousands of params trained on column representations

78 semantic types (name, address, etc)



Performance (F1)

Sherlock	0.89
Human Consensus	0.32



LLMs? Sherlock (*tiny* model) outperformed (*huge*) GPT-3.5 with +6%

Utility: from types to sensitivity?

Detecting Personal Information in Training Corpora: an Analysis

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Alexandra Sasha Luccioni *

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Jesse Dodge

AI2, USA

Margaret Mitchell

Hugging Face, USA

Abstract

Large language models are trained on increasing quantities of unstructured text, the largest sources of which are scraped from the Web. These Web scrapes are mainly composed of heterogeneous collections of text from multiple domains with minimal documentation. While some work has been done to identify and remove toxic, biased, or sexual language, the topic of *personal information* (PI) in textual data used for training Natural Language Processing (NLP) models is relatively underexplored. In this work, we draw from definitions of PI across multiple countries to define the first PI taxonomy of its kind, categorized by type and risk level. We then conduct a case study on the Colossal Clean Crawled Corpus (C4) and the Pile, to detect some of the highest-risk personal information, such as email addresses and credit card numbers, and examine the differences between automatic and

pretrained models to generate arbitrary sequences without modifying the underlying weights at all via steering vectors (Subramani et al., 2019; Subramani and Suresh, 2020; Subramani et al., 2022) and prompting (Shin et al., 2020; Li and Liang, 2021).

This suggests that it is necessary to better understand the types of PI contained in training corpora and the types of harms that they can cause, and to propose ways for automatically detecting (and, eventually, removing) the most high-risk types of PI from NLP corpora. We endeavor to address both of these directions in the current article: we start with defining different types of PI and propose a novel categorization in Section 2 and discuss the risks of different types of PI. Then, in Section 3, we explore the difficulty in detecting one of the highest-risk and easiest-to-identify types of PI, CHARACTER-BASED identifiers, comparing a

Are fixed “types” enough?

We argue:

sensitive data goes beyond types

“Data whose sensitivity depends on external factors and requires protection due to how, by whom, and in what context it can be misused.”

Data sensitivity is contextual!

🤔 Q1: are all PI typed columns, really sensitive?

personal address of human aid workers

Name	Role	Address	Region
Lina Haddad	Field Coordinator	14 Olive Grove Rd, Sector 3, Al Noor District	Northern Syria
David Mensah	Medical Logistics Officer	House 8, Kivu Relief Compound, East Ridge	Eastern DRC
Sara Petrov	Emergency Specialist	22 Cedar Lane, Block B, River Quarter	South Sudan

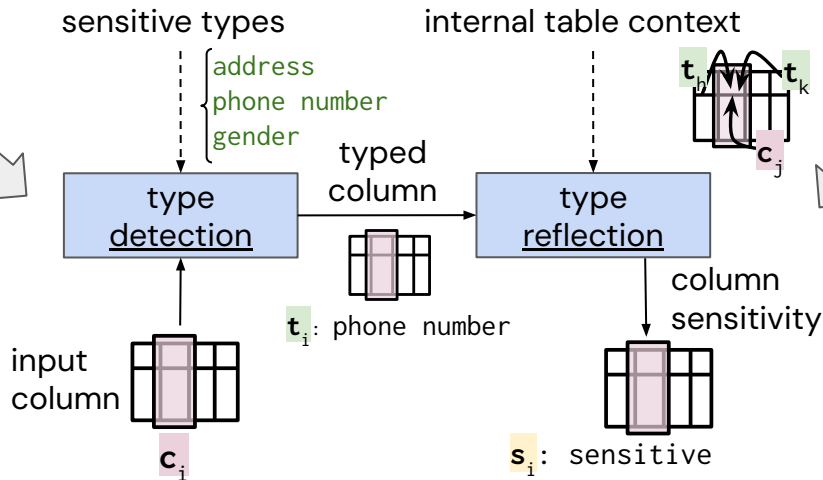
organizational address of professors

Name	Department	Address	Institution
Dr. Elena Vos	Sociology	Room 4.12, Social Sciences Building, 120 University Ave	Utrecht University
Prof. Michael Chen	Physics	Office 2-318, Science Hall, 85 Academic Way	TU Delft
Dr. Amina Rahman	History	Room 5B, Humanities Centre, 44 College Street	Leiden University

Mechanism: *type-contextualization*

w/o PII types:

recall -5%



w/o reflection:

precision -16%

✨ **integrated**
in UN's
humanitarian
data platform

94% sensitive PII detected
vs 63% with Google's DLP!

*Experiments with Qwen3 14B, on hand-labeled tables "in the wild" (GitTables)

Towards Contextual Sensitive Data Detection, Telkamp & Hulsebos, Findings of EMNLP, 2026.



CWI

Context beyond the table matters, too

🤔 Q2: are only PI typed columns, sensitive?

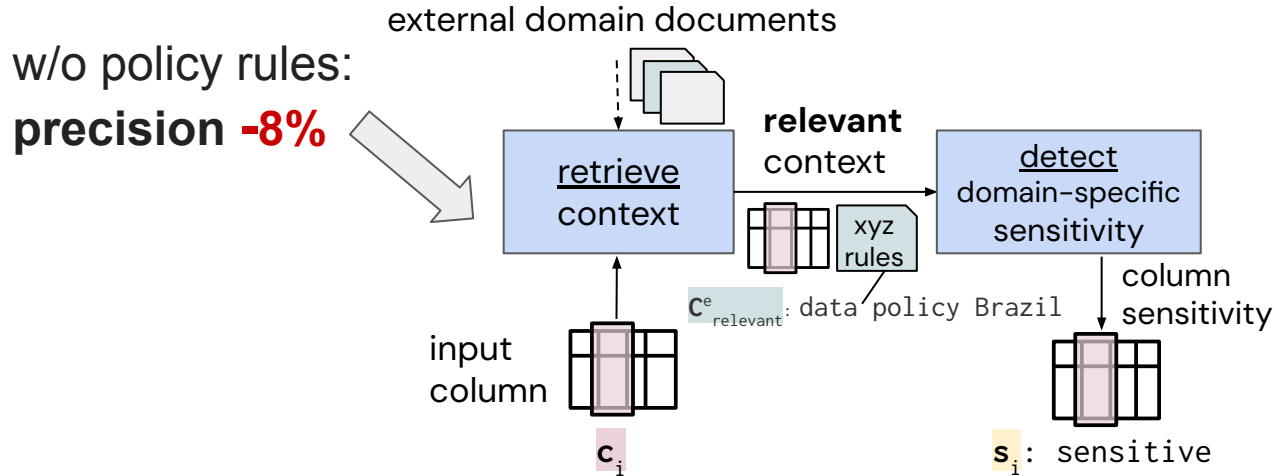
geo-coordinates from hospitals in **Gaza**

Facility Name	Lat	Long	Region
Al Amal Field Hospital	31.4872	34.4386	Central Gaza
Coastal Emergency Clinic	31.5129	34.4551	Northern Gaza
Rafah Community Hospital	31.2964	34.2478	Southern Gaza

geo-coordinates from hospitals in **Germany**

Facility Name	Lat	Long	Region
St. Marien Medical Center	52.5218	13.4097	Berlin
Rhine General Hospital	50.9386	6.9603	Cologne
Bavaria University Hospital	48.1397	11.5654	Munich

Mechanism: *domain-contextualization*



**100% sensitive non-PII data detected
but still yields some FPs**

*Generated and annotated by UN data experts

Case study: UN humanitarian data

INFORMATION SHARING PROTOCOL

The Occupied Palestinian Territory

25 April 2025

OVERVIEW

This Information Sharing Protocol (ISP) is designed to support *data responsibility* -- the *safe, ethical, and effective management of data* - within the Occupied Palestinian Territory (The OPT). It establishes a collective framework for information and data exchange on the basis of a common data and information sensitivity classification.

The OPT Assessment and Information Management Working Group (AIMWG) developed this ISP through a collective exercise in accordance with the Inter-Agency Standing Committee (IASC) Operational Guidance on Data Responsibility.¹ The ISP was endorsed by the OPT Inter-Cluster Coordination Group (ICCG) in June 2024 and revised by the AIMWG and ICCG the 26 and 28 February 2025

In this context, this ISP serves as the primary document governing data and information sharing in the humanitarian response in The OPT. This ISP should be used as a reference in the exchange of data and information with the Government of The OPT. This ISP complements existing policies and guidelines and does not in any way affect or replace obligations and restrictions contained in applicable legal and regulatory frameworks, Cluster-

High Sensitivity Information or data that, if disclosed or accessed without proper authorization, are likely to cause serious harm or negative impacts to affected people and/or humanitarian actors and/or damage to a response.	• Needs assessment data at the neighborhood (ADM4) level or lower • Survey results at the neighborhood (ADM4) level or lower • Explosive ordnance contamination data (exact locations or at ADM3 level) • Aid-worker contact details and contact lists if consent for sharing has not been obtained • Minutes of internal meetings • Information on evacuation of affected people or where local organizations are supporting relocation	Classification: Confidential Data or information can be disclosed within an organization or small community of organizations directly involved in delivering humanitarian assistance, based on a clearly specified purpose and related standards for data responsibility, including data protection. Methods for sharing confidential data: • Microsoft Teams files, SharePoint, OneDrive and similar collaboration platforms • Internal intra-Cluster sharing • Inter-Cluster sharing on case-by-case basis • HDX via HDX Connect • ArcGIS Online
Severe Sensitivity Information or data that, if disclosed or accessed without proper authorization, are likely to cause severe harm or negative impacts and/or damage to affected people and/or humanitarian actors and/or impede	• Humanitarian notification information • Raw survey data, such as individual survey responses, at the respondent or household level • Data on attacks on health facilities with locations (e.g. coordinates) • 5W data at the neighborhood (ADM4) level or site (ADM5) level • Live data on population movement based on metadata generated by electronic devices or based on	Classification: Strictly Confidential Highly limited, bilateral disclosure only. Determined and approved on a case-by-case basis, with assurance of upholding the highest standards of data responsibility, including data protection. Methods for sharing strictly confidential data:

Case study: UN humanitarian data

Reported location on NFIs suppliers								
nfi_supplier	nfi_supplier_loc- value -from_Iran	nfi_supplier	nfi_supplier	nfi_supplier	nfi_supplier	nfi_supplier	nfi_supplier	nfi_supplier
NA	0%	0%	0%	100%	0%	0%	0%	0%
NA	0%	0%	0%	33%	0%	67%	0%	0%
NA	0%	0%	0%	100%	0%	0%	0%	0%

Context-grounded LLM output

1. interpretation of data
2. assessment of sensitivity
3. context-grounded explanation

1. The column "nfi_supplier_loc- value -from_Iran" indicates the origin of NFIs suppliers.

2. This information, even with country-level disaggregation, doesn't reveal personal or sensitive data. It relates to the location of suppliers, which is generally not considered sensitive

3. It aligns with the ISP's guideline on "Facility data (e.g., health, education, water points) at national or regional level, unless explicitly restricted by clusters." Since this column is about locations of suppliers and not about individuals or specific operations, it is categorized as non-sensitive.

✨ UN Data Responsibility Lead: "I noticed the LLM explanation<s> and have to say it already looks great."

From *Semantic* to Analytical Reasoning

Analytical reasoning intends to produce **insights**

An **insight** is a collection of knowledge that links **analytical knowledge derived from data** with a person's **pre-existing contextual understanding**.

Type of Insights...

How to surface these insights?

Can one model serve all tasks?

Insight Types

Descriptive

Knowledge about the observed data, requiring no assumptions beyond it.

Lookup

Reading specific values or records present in the corpus, covering both attribute values for identified entities and sets of entities matching stated criteria.

"What was the population of Germany in 2022?"

"Which countries joined the European Union after 2000?"

Aggregation

Computing summary values over a set of records, including counts, sums, extrema, and rankings.

"Which Mercosur member country has the highest public debt as a share of GDP?"

Characterization

Describing the statistical distribution of a variable, including its central tendency, spread, and shape.

"How are household incomes distributed in New Zealand?"

Association

Measuring how two or more variables co-vary in the observed data, ranging from correlation to general dependence.

"What is the correlation between public education spending and literacy rates across countries?"

Grouping

Discovering segments of entities or records that share latent structure across attributes, without a predefined target variable.

"What clusters of countries exist based on their energy-mix profiles?"

Inferential

Knowledge about the data-generating process behind the observed data, requiring assumptions that link the two.

Statistical Inference

Determining whether an observed pattern reflects a stable feature of the underlying process rather than noise, with parameter estimates and quantified uncertainty.

"Is there evidence of a positive association between public education spending and literacy rates across countries?"

Prediction

Estimating unknown or future values for unobserved instances by generalizing observed patterns, spanning regression, classification, and forecasting.

"What is the expected population of Algeria in 2040?"

Causal

Knowledge about the effects of interventions or counterfactual states, requiring assumptions on the underlying causal structure.

Interventional

Determining the effect of an action or policy at the population level.

"How would commuting mode choices change if fuel prices rose by 25%?"

Counterfactual

Determining the outcome for a specific observed instance under a condition contrary to the known facts.

"Had the price not increased, would this customer have churned?"

Can we then just use LLMs for analytical querying?

What is the **average budget** of Drama movies with a **greater or equal to 8.0** rating?

MovieID	Title	MPAA Rating	Budget	Gross	Release Date	Genre	Runtime	Rating	Rating Count
121	Dangerous Minds	R	23 000 000	179 519 401	1995-08-11	Crime	99	6.5	45 479
377	Slumdog Millionaire	R	15 000 000	377 910 544	2008-05-12	Drama	120	8.0	758 215
386	Kung Fu Panda	PG	130 000 000	631 744 560	2008-06-04	Animation	90	7.5	390 478

ActorID	Name	Year of birth	Birth City	Birth Country	Height	Biography	Gender	Ethnicity	NetWorth
410	Dustin Hoffman	1937-08-08	LA	USA	65.0	"	Male	-	\$100M
553	George Dzundza	1945-07-19	Rosenheim	Germany	69.0	"	Male	White	\$2M
2099	Saurabh Shukla	1963-03-05	-	India	-	"	-	-	-

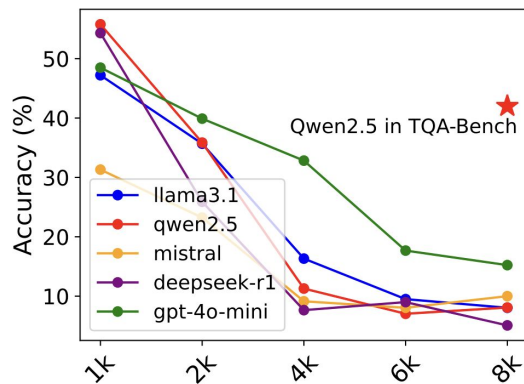
MovieID	ActorID	Character Name	creditOrder
121	553	Hal Griffith	2
377	2099	Sergeant Srinivas	2
386	410	Shifu	2

DeepSeek reasoning trace:

“... Wait no—the data doesn’t show that...
... Wait I’m getting confused...”

The expected sad state of tabular QA w/ LLMs directly

LLM-as-a-judge* on *averaging* questions
over multiple tables in TQA-Bench



let's not fool
ourselves with MPC
evals for LLMs
(source: TQA)

table context size (#tokens)

*>0.95% llm-as-a-judge accuracy (only 0.8% incorrect answers are judged as correct)

How well do LLMs reason over tabular data, really? by Wolff & Hulsebos, TRL@ACL, 2025

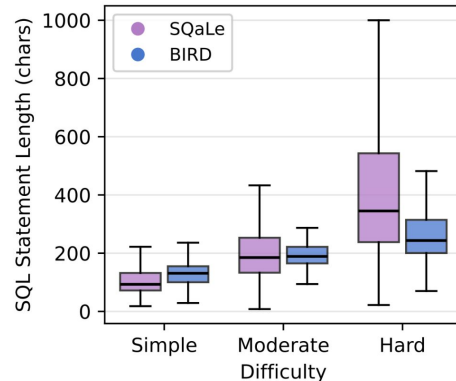
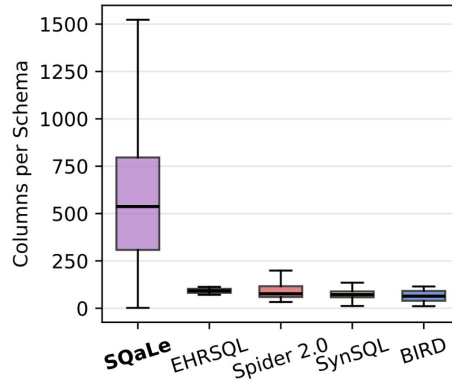
Better use SQL... (through LLM code gen)

Text-to-SQL is not entirely solved though:

- SOTA: throwing huge LLM pipelines at it? scalable? fit for “agentification”?
- Complex queries/schemas: text-to-SQL needs *data-grounding*

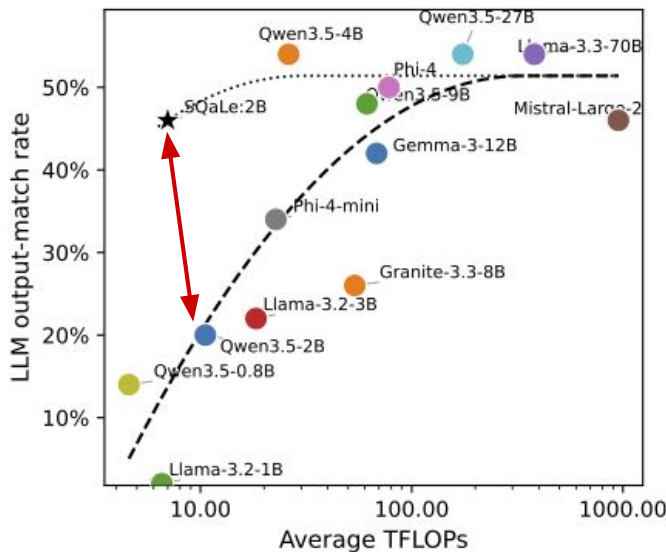
We need **SQaLe** to specialize.

- First **large** text-to-SQL dataset
- 700K+ (question, sql, schema)
- Queries over 6K **real** schemas (SchemaPile); large & diverse.



SQaLe enables the vision of *specialized small* text-to-SQL

Qwen-2B trained for agentic deployment using RL*



Training model on SQaLe is better than training on similar-sized “SynSQL”, as well as BIRD.

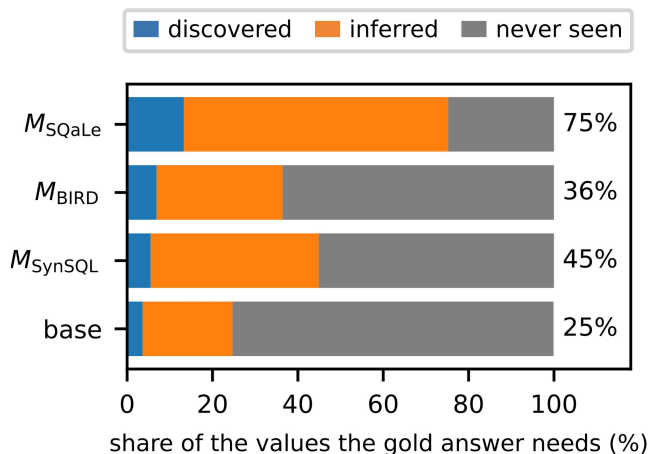
Why?

*On SQaLe test set

SQaLe: A Large Dataset for Specialized Text-to-SQL by Wolff, Gomm, and Hulsebos, in submission, 2026

A key insight: **agents need to learn to explore data!**

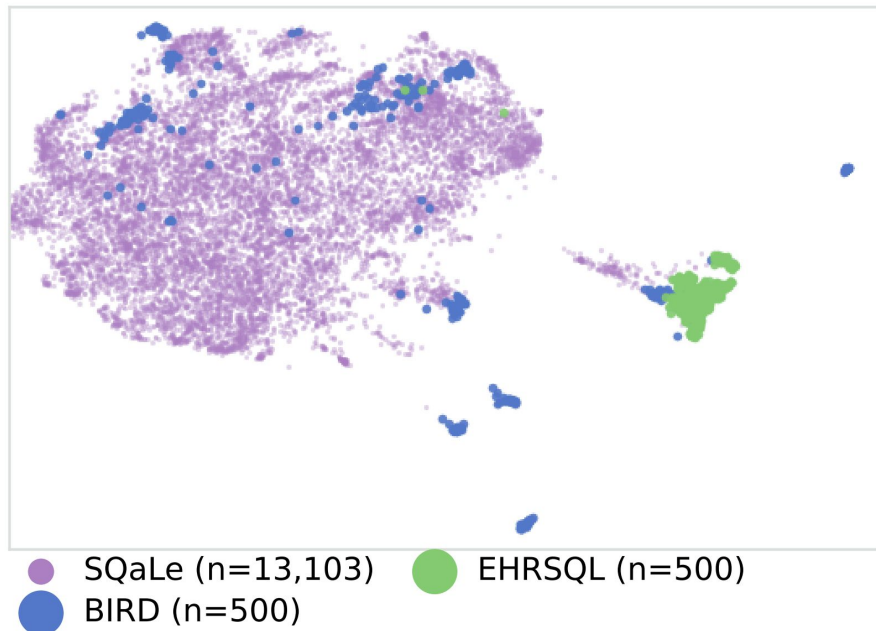
By training on **larger schemas** in #tables and #values (as in SQaLe), we observe ***emergent “schema exploration” behavior in text-to-SQL agent***



M_{SynSQL} makes most tool calls yet recovers relevant values for only 45%.

M_{SQaLe} recovers the relevant values for 75%!

Also: diversity in SQaLe



Generalization from coverage!

Domain	M_{SQaLe}	M_{BIRD}	Diff.
In-domain	39.3	36.5	+2.8
Out-of-domain	46.4	36.5	+9.9

Complex queries in SQALe, you said?

An example question in **SQALe**:

"For each customer in the gold or platinum loyalty tier who has placed at least two delivered orders in the last six months, return their full name, email, city and state, the total number of delivered orders, the total amount they paid based only on captured payments, their most frequently ordered product category in that period, and whether any of the products they ordered currently have low inventory in any warehouse (defined as quantity on hand minus quantity reserved less than 10), sorted by total captured payment amount from highest to lowest."



```
WITH gold_platinum_customers AS
(SELECT c.id,
       c.first_name,
       c.last_name,
       c.email,
       c.city,
       c.state
FROM customers c
JOIN orders o ON c.id = o.customer_id
WHERE c.loyalty_tier IN ('gold',
                        'platinum')
AND o.status = 'delivered'
AND o.delivered_at >= date('now', '-6 months')
GROUP BY c.id
HAVING COUNT(*) >= 2),
order_details AS
(SELECT gpc.id,
       gpc.first_name,
       gpc.last_name,
       gpc.email,
       gpc.city,
       gpc.state,
       o.order_number,
       SUM(p.amount) AS total_paid,
       p.status AS payment_status
FROM gold_platinum_customers gpc
JOIN orders o ON gpc.id = o.customer_id
LEFT JOIN payments p ON o.id = p.order_id
AND p.status = 'captured'
WHERE o.delivered_at >= date('now', '-6 months')
AND o.status IN ('delivered',
                 'completed'))
GROUP BY gpc.id,
       o.id,
       product_category AS
(SELECT od.id,
       od.first_name,
       od.last_name,
       od.email,
       od.city,
       od.state,
       p.category,
       ROW_NUMBER() OVER (PARTITION BY od.id
                        ORDER BY COUNT(*) DESC) AS rn
FROM order_details od
JOIN order_items oi ON od.order_number = CAST(oi.order_id AS TEXT)
JOIN products p ON oi.product_id = p.id
GROUP BY od.id)
SELECT gpc.first_name,
       gpc.last_name,
       gpc.email,
       gpc.city,
       gpc.state,
       COUNT(od.order_number) AS total_orders,
       SUM(od.total_paid) AS total_captured_payment_amount
FROM gold_platinum_customers gpc
JOIN order_details od ON gpc.id = od.id
AND od.payment_status = 'captured'
LEFT JOIN product_category pc ON od.id = pc.id
AND pc.rn = 1
WHERE pc.category IS NOT NULL
GROUP BY gpc.id
ORDER BY total_captured_payment_amount DESC;
```

*SQALe has different verbosity levels :)

From *Analytical* to *Inferential* Reasoning

Flashy demos of Data Science Agents

 Data Science Agent Experiment

 README.md  Playground

2018_Central_Park_Squirrel_Census_-_Squirrel_Data_20240501.csv

Analyze the proportion of adult and juvenile animals in the census data. Are there any spatial patterns in age distribution?

2018_Central_Park_Squirrel_Census_-_Squirrel_Data_20240501.csv

Plan

Problem Solved?

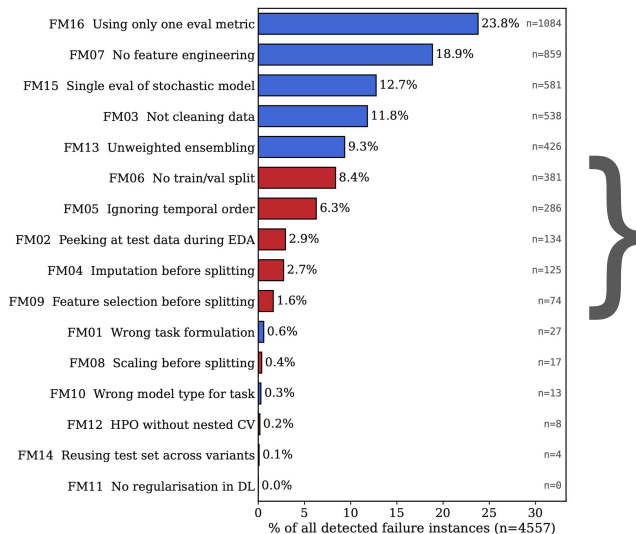
Model	SDE (69 tasks)		PM (28 tasks)		DS (14 tasks)	
	Success	Score	Success	Score	Success	Score
Closed model APIs						
Claude-3.5-Sonnet	30.43	38.02	35.71	51.31	14.29	21.70
Gemini-2.0-Flash	13.04	18.99	17.86	31.71	0.00	6.49
GPT-4o	13.04	19.18	17.86	32.27	0.00	4.70
Gemini-1.5-Pro	4.35	5.64	3.57	13.19	0.00	4.82
Amazon-Nova-Pro-v1	2.90	6.07	3.57	12.54	0.00	3.27
Open-weight models						
Llama-3.1-405b	5.80	11.33	21.43	35.62	0.00	5.42
Llama-3.3-70b	11.59	16.49	7.14	19.83	0.00	4.70
Qwen-2.5-72b	7.25	11.99	10.71	22.90	0.00	5.42
Llama-3.1-70b	1.45	4.77	3.57	15.16	0.00	5.42
Qwen-2-72b	2.90	3.68	0.00	7.44	0.00	4.70

data science is
not just
code generation

Just LLMing won't do it, but just SQLing neither...

We're studying Failure Modes in agentic data science

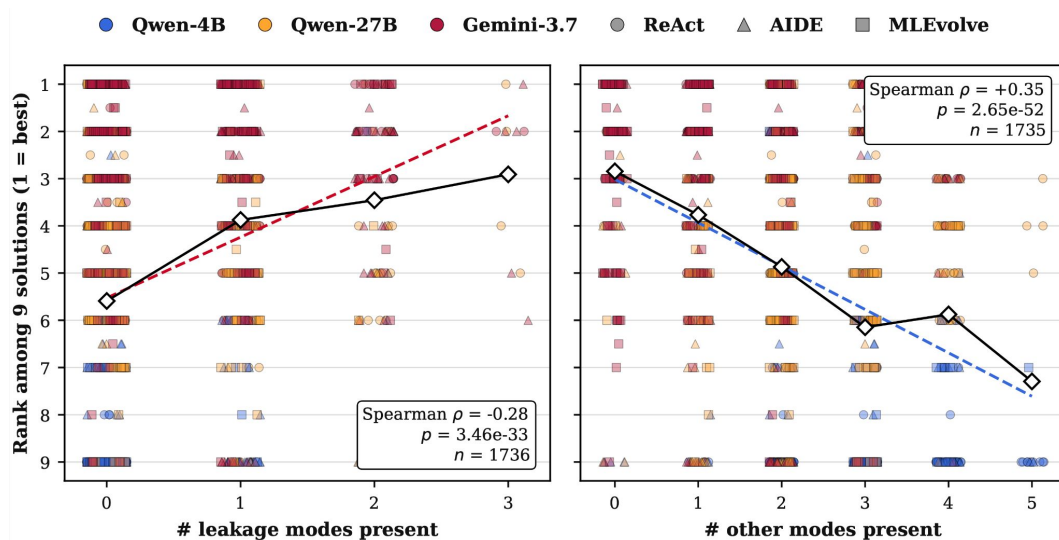
Distribution of failure modes across **220 DS pipelines**, 3 DS agents & 3 different models



Besides common DS FMs, we found many flavors of leakage.

Looking Beyond Accuracy: Why do Data Science Agents fail? by Li, X., et al, in submission, 2026.

Stop evaluating only downstream performance!



Tentative results:
work in progress!

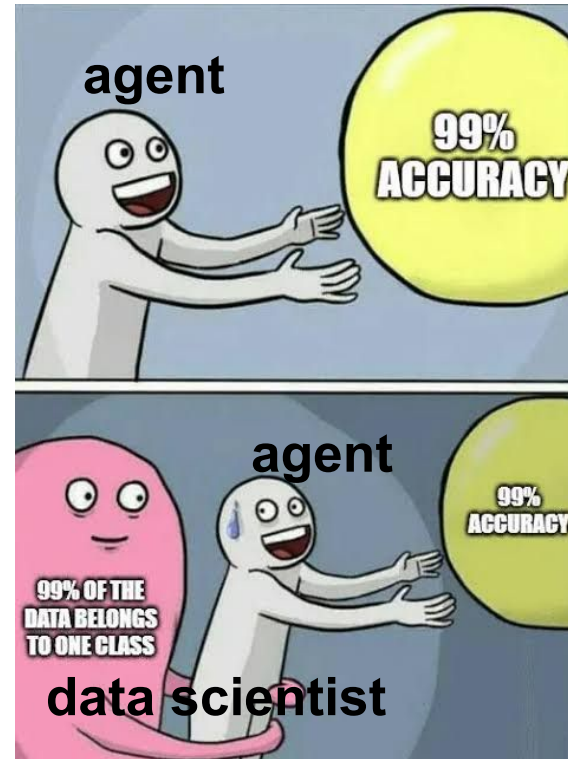
“cheating” models == better performance

Looking Beyond Accuracy: Why do Data Science Agents fail? by Li, X., et al, in submission, 2026.

A Happy End...

Agents clean the data...

Data scientists do the creative
and rigorous thinking...



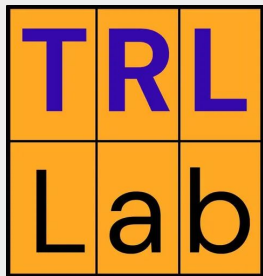
Recap on the Flavors of Tabular Reasoning

Wrapping up...

→ Tabular AI is Exciting!!!

- For tasks powered by table semantics (e.g. sensitivity val), contextualization is key for LLMs for improving, interpreting, and consistency of outcomes.
- Tabular AI is **multifaceted**, no future for “general-purpose (tabular) model”: **semantic / analytical / statistical reasoning are different in nature**. Future for small specialized models, text2sql, retrieval, prediction...
- Needs of *agentic data science* for predictive tasks go beyond contextualized LLMs/code gen: domain- and data science knowledge are key. We need human expertise here... !

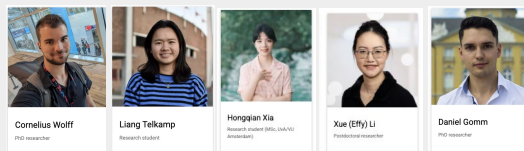
That's it!



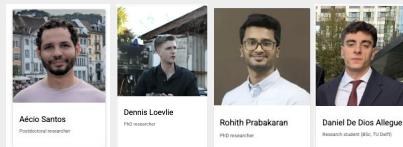
`trl-lab.github.io`

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> **maelon@cw.nl**

Leads of projects



Other members



Funding & Collaboration

